2.9.1 Properties of Logarithm

(i) $a^{\log_a x} = x$ for all $x \in (0, \infty)$ and $\log_a (a^y) = y$ for all $y \in \mathbb{R}$.

(ii) For any $x, y > 0$, $\log_a(xy) = \log_a x + \log_a y$. (Product Rule)

(iii) For any $x, y > 0$, $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$. (Quotient Rule)

(iv) For any $x > 0$ and $r \in \mathbb{R}$, $\log_a x^r = r \log_a x$. (Power Rule)

(v) For any $x > 0$, with a and b as bases, $\log_b x = \frac{\log_a x}{\log_a b}$. (Change of base formula.)

Proof. Since exponential function with base a and logarithm function with base a are inverse of each other,

(i) follows by using the definitions.

(ii) For $x, y > 0$ let $\log_a x = u$, $\log_a y = v$, and $\log_a(xy) = w$. Rewriting these in the exponential form we obtain $a^u = x$, $a^v = y$, and, $a^w = xy$. So, $a^w = xy = a^u a^v = a^{u+v}$; thus $w = u + v$. Thus, we obtain $\log_a(xy) = \log_a x + \log_a y$.

(iii) Let $\log_a x = u$, $\log_a y = v$, and $\log_a \frac{x}{y} = w$. Then $a^u = x$, $a^v = y$ and $a^w = \frac{x}{y}$. Hence, $a^w = \frac{x}{y} = \frac{a^u}{a^v} = a^{u-v}$; which implies $w = u - v$.

Thus, we obtain $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$.

(iv) Let $\log_a x = u$. Then $a^u = x$ and therefore, $x^r = (a^u)^r = a^{ru}$. Thus, $\log_a x^r = ru = r \log_a x$.

(v) Let $\log_b x = v$. We have $b^v = x$. Taking logarithm with base a on both sides we get $\log_a b^v = \log_a x$.

On the other hand $\log_a b^v = v \log_a b$ by the Power rule. Therefore, $v \log_a b = \log_a x$.
Hence $\log_b x = \frac{\log_a x}{\log_a b}$, $b > 0$. This completes the proof. ∎

Remark:

(i) If $a = 10$, then the corresponding logarithmic function $\log_{10} x$ is called the **common logarithm**.

(ii) If $a = e$,(an irrational number, approximately equal to 2.718), then the corresponding logarithmic function $\log_e x$ is called the **natural logarithm**. It is denoted by $\ln x$. These above particular cases of logarithmic functions are used very much in other sciences and engineering. Particularly, the natural logarithm occurs very naturally. When we write $\log x$ we mean $\log_e x$.

(iii) If $a = 2$, then the corresponding logarithmic function $\log_2 x$ called the binary logarithm, which is used in computer science.

(iv) Observe that $\log_a 35 = \log_a(7 * 5) = \log_a 7 + \log_a 5$; $\log_a \frac{50}{3} = \log_a 50 - \log_a 3$.

$\log_a 22^x = x \log_a 22$; $\log_5 50 = \frac{\log_{10} 50}{\log_{10} 5}$.

(v) Observe the graph of the logarithmic and exponential functions.